



Project Title

Chatbot for College Enquiry System using Natural Language Processing (NLP)

Abstract

The proposed project aims to develop an intelligent chatbot that can automatically answer college-related queries such as admissions, courses, fee structure, eligibility, scholarships, hostel facilities, placements, and examination schedules. The chatbot will use Natural Language Processing (NLP) techniques to understand user queries and provide relevant responses. This system is expected to reduce the workload of administrative staff while offering students quick and accurate information. The project demonstrates the practical application of Data Science and Machine Learning in educational institutions.

Introduction

Educational institutions receive a large number of enquiries from students and parents, especially during admission periods. Most of these questions are repetitive and require administrative staff to provide the same information multiple times. This process is time-consuming and may lead to delays in responding to users.

The proposed chatbot will automate this process by understanding user queries written in natural language and providing instant responses. By applying NLP and Machine Learning techniques, the chatbot will improve communication efficiency and provide students with information anytime without human assistance.

Problem Statement

Traditional enquiry systems rely on manual communication through phone calls, emails, or office visits. These methods consume time and increase the workload of staff. Static FAQ pages also fail to understand different ways in which users ask



questions. Therefore, an intelligent chatbot is required to understand user intent and provide accurate responses automatically.

Objectives

The main objective of this project is to develop an intelligent chatbot that can answer common college-related enquiries using Natural Language Processing. The system aims to provide quick responses, reduce manual work, improve user experience, and demonstrate the application of Data Science techniques such as text preprocessing and machine learning.

Proposed Methodology

The chatbot will be developed using Python and NLP techniques. A dataset of frequently asked college-related questions and answers will be collected from official college websites and admission brochures. The textual data will be preprocessed using tokenization, stop-word removal, and lemmatization. TF-IDF will be used for feature extraction, while machine learning algorithms such as Naïve Bayes or Logistic Regression will classify user intent. Based on the predicted intent, the chatbot will return the most relevant response.



Software and Hardware Requirements

The project will use Python, Pandas, NumPy, NLTK or spaCy, Scikit-learn, and Flask or Streamlit for development. A computer with at least an Intel Core i3 processor, 8 GB RAM, and 10 GB free storage will be sufficient for implementation.

Expected Outcome

The proposed chatbot is expected to provide accurate answers to common college enquiries, reduce administrative workload, improve response time, and enhance student satisfaction. The project will also demonstrate the practical implementation of NLP and Machine Learning techniques in solving real-world educational problems.

Future Scope

The chatbot can be enhanced by integrating advanced AI models, multilingual support, voice-based interaction, WhatsApp integration, and direct connection with the college ERP system. These improvements will make the chatbot more intelligent and user-friendly.

Conclusion

The proposed Chatbot for College Enquiry System is a useful Data Science project that applies Natural Language Processing and Machine Learning to automate student enquiry services. The system is expected to improve communication, save time, and



provide accurate information efficiently. It also offers a strong foundation for future AI-based educational applications.

References

1. Jurafsky, D., & Martin, J. H. *Speech and Language Processing*.
2. NLTK Documentation.
3. Scikit-learn Documentation.
4. Research papers on NLP Chatbots.